

Daily Work Report

June 19 — July 1, 2026

Date	Day	Focus Area	Key Output
Jun 19	Fri	Backtest	Identified top performing strategies: EngulfRaw_CH55 with MA...
Jun 20	Sat	Backtest	Options 1L: 55-60% win rate with 1.5-2x RR. BB Breakout: 62%...
Jun 21	Sun	Options	Same-day TP35 win rate improved to 65% with 1.2x avg RR. SL ...
Jun 22	Mon	Data	Multi-day TP=150, D=14: 58% win rate. SL sweeps identified o...
Jun 23	Tue	Model	Return prediction v1: AUC=0.62. Directional accuracy=53%. To...
Jun 24	Wed	Pipeline	V5 ensemble (XGB+LGB+RF): AUC=0.634. Gap fade: 58% win rate,...
Jun 25	Thu	Model	V7 walkforward: AUC=0.628 [0.622-0.634]. Champion model (85 ...
Jun 26	Fri	Pattern	60-min MTF: 6.9M predictions, 477 symbols, corr=0.022. High ...
Jun 27	Sat	Pipeline	35 consensus features selected (9 methods). AUC progression:...
Jun 28	Sun	Pipeline	Hyperparameter tuning completed. Best params: learning_rate=...
Jun 29	Mon	Model	PROJECT COMPLETED. Best AUC=0.6545 (Phase 8 baseline). Final...
Jul 1	Wed	Model	Model parameters and feature importance archived....

Jun 19 (Friday)

Backtest Strategy Development — Engulfing patterns, MA directional filters, sweep optimization. Generated Comprehensive Strategy Report, Monthly/Yearly Visualization, Best Strategy Viz.

File / Output	Description
check_gaps_fast.py	Initial gap checking in strategy data
check_all_strategies.py	Audit all strategy variants for correctness
deep_research_v2.py	Deep research on strategy patterns
agentic_system_v2.py	Agentic system for automated backtest
root_cause_analysis.py	Root cause analysis of losses
fix_strategy.py / optimal_fix.py	Fix identified strategy bugs
final_fixed_test.py	Final validation of fixed strategies
test_improvements.py	Test improvement ideas
load_delivery.py	Load delivery data for analysis
full_history_test.py	Full history backtest validation
verified_backtest.py	Walkforward verified backtest
all_improvements_combined.py	Combine all strategy improvements
deep_stats_analysis.py / v2.py	Deep statistical analysis of strategies
final_consolidated_ranking.py	Rank all strategies by performance
generate_report_pdf.py	Generate Backtest_Report.pdf
engulfing_all_versions.py	All engulfing candle pattern versions
monthly_yearly_all.py	Monthly/yearly performance breakdown
loss_root_cause.py / pattern_mining.py	Mine patterns from losses
gen_work_report_pdf.py	Generate work_report_dates.pdf
Complete_Project_History.pdf	Complete project history report (19.3 KB)
Backtest_Report.pdf	Strategy backtest report (21.3 KB)
Comprehensive_Strategy_Report.pdf	Comprehensive strategy analysis (34.1 KB)
Monthly_Yearly_Visualization.pdf	Monthly/yearly performance viz (133 KB)
Best_Strategy_Viz_EngulfRaw_CH55.pdf	Best strategy visualization (121 KB)
Complete_Project_History.pdf	Complete project history report
work_report.pdf	Work report (16.7 KB)

Result: Identified top performing strategies: EngulfRaw_CH55 with MA directional filter. Generated 8 PDF reports documenting all findings.

Jun 20 (Saturday)

Options Backtesting & Data Download — Options 1L backtest, Nifty 500 bulk download, Bollinger Band breakout strategy, SENSEX/BANKNIFTY 1L backtests, data dedup/quality.

File / Output	Description
option_backtest.py / v2.py / v3.py	Options backtest framework (3 iterations)
option_all_variants.py	All option strategy variants
option_deep_itm.py	Deep ITM option analysis
option_exit_sweep.py	Exit strategy sweep for options
option_combined.py	Combined option strategy
option_1l_backtest.py	1-lot option backtest
option_fixed_backtest.py	Fixed option engine
option_pdf_report.py	Option Exit Sweep Report PDF
sensex_banknifty_backtest.py	SENSEX + BANKNIFTY 1L backtest
backtest_bb_breakout.py	Bollinger Band breakout backtest
sweep_bb_strategies.py	Sweep all BB strategy parameters
download_nifty500_batch.py	Download Nifty 500 stock data
dl_1min.py	Download 1-min historical data
dedup_compare.py / clean_data_dedup.py	Deduplicate and clean data
data_quality_check.py	Full data quality audit
tradebook_analysis.py	Tradebook performance analysis
same_day_audit.py / same_day_final.py	Same-day option final results
perf_analysis_tradebook.pdf	Tradebook performance analysis (338 KB)
FULL_REPORT.pdf	Full options research report (265 KB)
Option_Exit_Sweep_Report.pdf	Options exit sweep (119 KB)
Option_1L_Backtest_Report.pdf	1-lot options backtest (126 KB)
SENSEX_1L_Backtest_Report.pdf	SENSEX 1-lot backtest (224 KB)
BANKNIFTY_1L_Backtest_Report.pdf	BANKNIFTY 1-lot backtest (220 KB)
BB_Breakout_Strategy_Report.pdf	BB breakout report (10.9 KB)
FINAL_TP75_Max10d_Report.pdf	Final TP75 Max10d (107 KB)
SameDay_TP75_EOD1525_Report.pdf	Same-day TP75 report (82 KB)
SameDay_MD_TP200_Max10d_Report.pdf	Same-day MD report (86 KB)
trade_book.pdf	Full trade book (2.9 MB)
2pct_vs_5pct_comparison.pdf	2% vs 5% comparison (110 KB)
monthly_analysis_2pct.pdf	Monthly analysis 2% (346 KB)

Result: Options 1L: 55-60% win rate with 1.5-2x RR. BB Breakout: 62% win rate. Downloaded ~200 Nifty 500 stocks. Generated 15+ PDF reports.

Jun 21 (Sunday)

Same-Day Options Optimization — Optimized same-day strategy parameters (TP=35, Mon entry), refined SL optimization.

File / Output	Description
same_day_optimized.py	Optimized same-day strategy parameters
final_tp35_mon.py	TP=35 Monday entry final version
same_day_refine.py	Refined same-day parameters
final_analysis_pdf.py	Full same-day analysis PDF
sl_optimize.py	Stop-loss optimization sweep
SameDayOpt_TP35_Report.pdf	Optimized TP35 report (49 KB)

SameDay_TP35_Mon_Entry12.pdf	TP35 Mon entry12 (53 KB)
SameDay_FullAnalysis.pdf	Full same-day analysis (86 KB)

Result: Same-day TP35 win rate improved to 65% with 1.2x avg RR. SL optimization found optimal SL=15 for best risk-adjusted returns.

Jun 22 (Monday)

Multi-Day Strategies & Data Filling — Multi-day hold strategies, expiry gap filling, stop-loss sweeps across all timeframes, comprehensive reporting.

File / Output	Description
multi_day_check.py / multi_day_pdf_report.py	Multi-day hold strategy backtest
md_tp150_d14.py	Multi-day TP=150, hold 14 days
check_expiry_gap.py / check_hold_expiry.py	Expiry gap analysis for multi-day
fill_data_gaps.py / verify_filled.py	Fill gaps in historical data
final_comprehensive_sweep.py	Comprehensive parameter sweep
sl_sweep_full.py / sl_sweep_15min.py	Stop-loss sweeps (full + 15min)
all_timeframes_sweep.py	All timeframe strategy sweep
comprehensive_report.py / timeframes_report.py	Generate PDF reports
MultiDay_FullReport_FIXED.pdf	Multi-day fixed report (149 KB)
FilledData_Report.pdf	Filled data report (109 KB)
Comprehensive_Report.pdf	Comprehensive report (2.7 MB)
Timeframes_Report.pdf	All timeframes report (947 KB)
model_architecture_full_report.pdf	Model architecture report (337 KB)

Result: Multi-day TP=150, D=14: 58% win rate. SL sweeps identified optimal SL=12 for 5min, SL=20 for 15min. Data gaps filled for 47 stocks.

Jun 23 (Tuesday)

Return Prediction Model v1 — Feature analysis report, return prediction model (XGB/LGB/RF ensemble), directional backtest, tradebook generation.

File / Output	Description
FEATURE_ANALYSIS_REPORT.md/.pdf	Feature analysis of 50+ indicators (1 MB PDF)
RETURN_PREDICTION_REPORT.md/.pdf	Return prediction model v1 (207 KB)
directional_backtest_report.pdf	Directional backtest (410 KB)
pred_vs_actual_report.pdf	Predicted vs actual analysis (189 KB)
return_model_tradebook.pdf	Return model tradebook (12.5 KB)
results.pkl	V1 results (172 KB)
feature_data.parquet	Feature dataset (104 MB)

Result: Return prediction v1: AUC=0.62. Directional accuracy=53%. Top features: MACD, RSI, Volume ratios. 446 symbols, 89K rows.

Jun 24 (Wednesday)

Data Pipeline + V3/V4/V5 Models — Data pipeline debugging, V3/V4 prediction training, gap fade intraday strategy, v5 ensemble model, forensic audit.

File / Output	Description
results_v2.pkl	V2 results (5.9 MB)
results_v3.pkl	V3 results (21.6 MB)
gap_fade_full.py / fast.py	Gap fade intraday strategy backtest
generate_report_v5.py	V5 ensemble model report
v5_report.pdf	V5 report (695 KB)
debug_open.py / debug_lookup.py	Debug open price issues

trade_book_v5.py	V5 trade book generation
v5_trade_book.pdf	V5 trade book (495 KB)
audit_v5.py / forensic_audit_*.py	Full forensic audit (Phase 2,3,6,7)
regime.py / indicators.py / fix_bb_pct_b.py	Feature computation fixes
results_raw.pkl	Raw results (35 MB)
results_v6.pkl	V6 results (62.6 MB)
train_v6.py	V6 model training

Result: V5 ensemble (XGB+LGB+RF): AUC=0.634. Gap fade: 58% win rate, 1.3x avg return. V6 with 220 features: AUC=0.641. Forensic audit completed for Phases 2,3,6,7.

Jun 25 (Thursday)

V6/V7 Models + Champion Model — V6 training with 220 features, V7 with purged walkforward, champion model ensemble, 60min/MTF/directional models.

File / Output	Description
train_v7.py / train_v7_from_pickle.py	V7 model with walkforward validation
models_v7.pkl	V7 saved models (24.7 MB)
results_v7.pkl	V7 results (132 MB)
results_v5.pkl	V5 final results (128 MB)
results_v6.pkl	V6 updated results (64 MB)
champion_model.py	Champion ensemble model
results.pkl (champion)	Champion model results (68 MB)
directional_model_v2.py	Directional (up/down) classifier v2
strategy_60min.py / total.py	60-min strategy models
strategy_mtf_60min.py / mtf_daily.py	Multi-timeframe strategy
results_60min.pkl	60-min results (52 MB)
strategy_weekly_factor.py	Weekly factor model
training_data.pkl	Training data export (1.75 GB)

Result: V7 walkforward: AUC=0.628 [0.622-0.634]. Champion model (85 features): lift=1.82x at th=0.55. Directional: AUC=0.512. 60-min MTF: R2=-0.03. Best recall=57.3% (cluster-specific).

Jun 26 (Friday)

Pattern Mining + 60min MTF + High Gainer Classifier — Pattern mining v2, daily/60-min MTF strategies, high gainer classifier (XGB/LGB).

File / Output	Description
mining_v2.py / backtest_rules.py	Pattern mining v2
strategy_daily_mtf.py	Daily MTF strategy
results_daily_mtf.pkl	Daily MTF results (30 MB)
strategy_60min_mtf.py	60-min MTF strategy (44 features)
results_60min_mtf.pkl	60-min MTF results (239 MB)
strategy_daily_classifier.py	High gainer daily classifier
classifier_results.pkl	Classifier results (45 MB)
strategy_hg_classifier_v2.py	Classifier v2
audit_quality.py / fix_quality_issues.py	Data quality audit + fix

Result: 60-min MTF: 6.9M predictions, 477 symbols, corr=0.022. High gainer classifier: AUC=0.654 (best single model). Pattern mining: 47 patterns found, 65% win rate on top 5.

Jun 27 (Saturday)

Phase 4-8 Full Pipeline — Feature engineering (157 features), data cleaning, EDA, time series analysis, feature selection (9-method consensus), improvement pipeline.

File / Output	Description
phase7_feature_engineering.py	Phase 7: 157 engineered features
engineered_features.parquet	Engineered features (294 MB)
phase4_data_mining.py	Phase 4: data mining
phase5_eda.py / fix / tail	Phase 5: EDA
phase6_data_cleaning.py	Phase 6: data cleaning
cleaned_features.parquet	Cleaned dataset (386 MB, 1M rows)
phase_tsa_time_series_analysis.py	Time series analysis
deep_comprehensive_analysis.py	Deep comprehensive analysis
deep_tsa_only.py	Time series analysis only
gen_comprehensive_report.py	Comprehensive report PDF
COMPREHENSIVE_REPORT.pdf	Comprehensive report (1.5 MB)
gen_vol_report.py / vol_breakout.py	Volatility breakout analysis
vol_breakout_report.pdf	Vol breakout report (521 KB)
phase6_depth_analysis.py	Phase 6 depth analysis
PHASE6_DATA_QUALITY_REPORT.pdf	Phase 6 report (551 KB)
phase8_feature_selection.py	Phase 8: 35 features from 109 via consensus
PHASE8_FEATURE_SELECTION_REPORT.pdf	Phase 8 report (669 KB)
phase8_evaluate.py	Phase 8 evaluation
improved_features.parquet	Improved dataset (355 MB)
IMPROVEMENT_REPORT.pdf	Improvement report (5 KB)
phase9_walkforward.py	Phase 9: walkforward fold definitions

Result: 35 consensus features selected (9 methods). AUC progression: 0.589 (1 feat) → 0.614 (8 feats) → 0.6545 (35 feats). Class weights > SMOTE. 1M rows × 116 cols cleaned.

Jun 28 (Sunday)

Phase 10-14 Tuning — Hyperparameter tuning, final evaluation, cluster-specific models.

File / Output	Description
phase10_14_tuning.py	Phase 10-14: tuning + evaluation pipeline

Result: Hyperparameter tuning completed. Best params: learning_rate=0.03, n_estimators=500, num_leaves=63. Cluster models tested (8 vol×mcap clusters).

Jun 29 (Monday)

Final Model & Report — Final cluster-enhanced ensemble, AGENTS.md, FINAL_PROJECT_REPORT.pdf.

File / Output	Description
final_eval.py	Final class-weights evaluation
final_model.py	Cluster-enhanced ensemble model
final_model_results	229 features, AUC=0.628, 95%CI [0.622-0.634]
AGENTS.md	Project summary with key results
final_report_pdf.py	Generate FINAL_PROJECT_REPORT.pdf
FINAL_PROJECT_REPORT.pdf	Final project report (9.5 KB)

Result: PROJECT COMPLETED. Best AUC=0.6545 (Phase 8 baseline). Final ensemble AUC=0.628. Key finding: cross-sectional > single-stock. AUC ceiling ~0.655 with price/volume data alone.

Jul 1 (Wednesday)

Model maintenance — LGB model text dump for inspection.

File / Output	Description
lgb_model.txt	LGB model text dump (19.5 KB)

Result: Model parameters and feature importance archived.

Overall Statistics

Period: June 19 — July 1, 2026 (13 days)

Total files created: 200+ (scripts, models, reports, datasets)

Best model result: AUC=0.6545 (Phase 8 baseline, 35 features, LightGBM class weights)

Final ensemble: AUC=0.628 [95%CI: 0.622-0.634] (cluster-enhanced, 229 features)

Key insight: Cross-sectional ranking > single-stock timing. AUC ceiling ~0.655 with price/volume.

PDF reports generated: 40+

Report generated: 2026-07-02 10:59